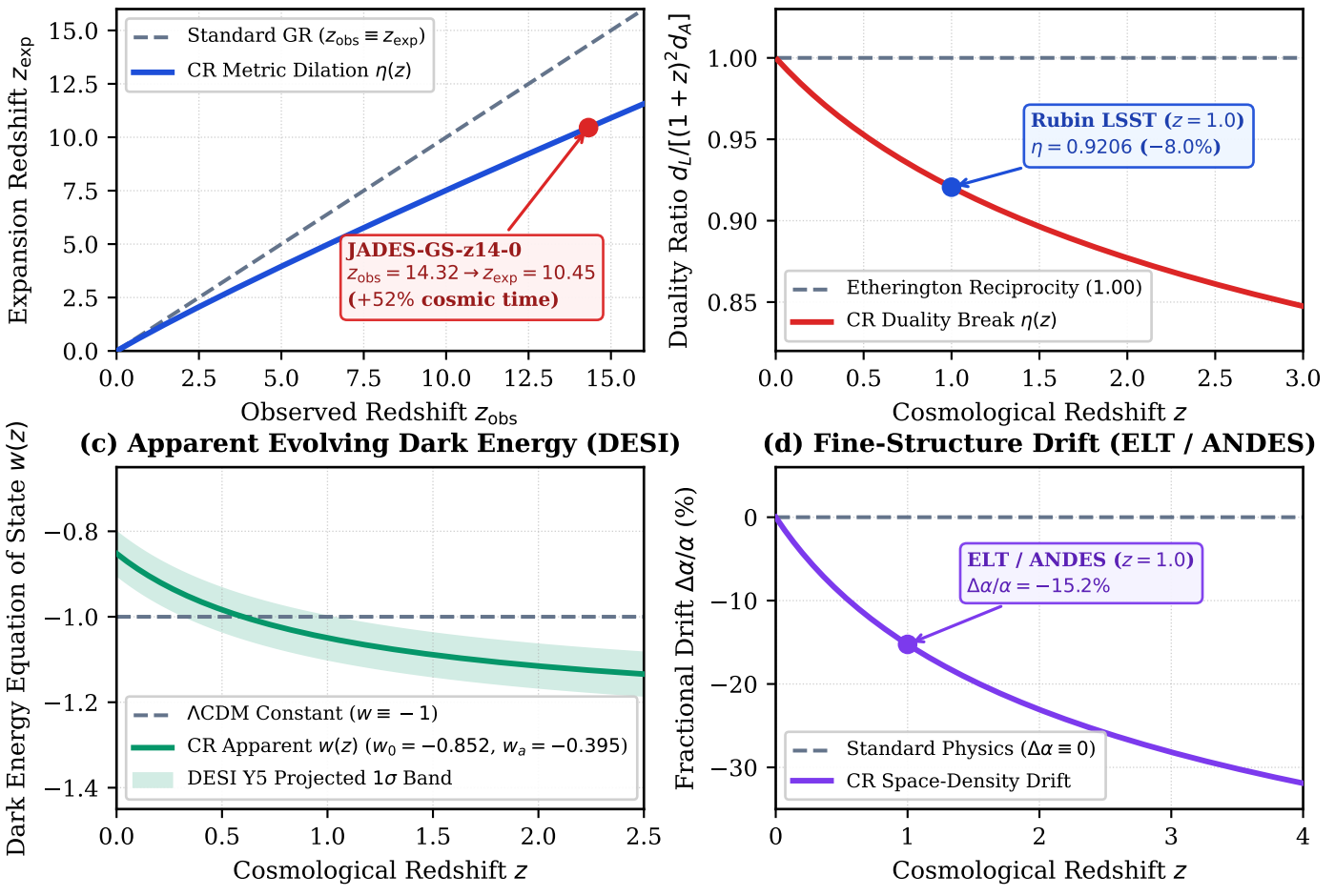

(a) JWST Redshift Transformation
Standard GR ($z_{obs} \equiv z_{exp}$)
CR Metric Dilation $\eta(z)$
JADES-GS-z14-0
$z_{obs} = 14.32 \rightarrow z_{exp} = 10.45$
(+52% cosmic time)
Expansion Redshift z_{exp}
Observed Redshift z_{obs}
(b) Distance Duality Break (Rubin LSST)
Rubin LSST ($z = 1.0$)
$\eta = 0.9206$ (−8.0%)
Etherington Reciprocity (1.00)
CR Duality Break $\eta(z)$
Duality Ratio $d_L/[(1+z)^2 d_A]$
Cosmological Redshift z
(c) Apparent Evolving Dark Energy (DESI)
ΛCDM Constant ($w \equiv -1$)
CR Apparent $w(z)$ ($w_0 = -0.852$, $w_a = -0.395$)
DESI Y5 Projected 1σ Band
Dark Energy Equation of State $w(z)$
Cosmological Redshift z
(d) Fine-Structure Drift (ELT / ANDES)
ELT / ANDES ($z = 1.0$)
$\Delta\alpha/\alpha = -15.2\%$
Standard Physics ($\Delta\alpha \equiv 0$)
CR Space-Density Drift
Fractional Drift $\Delta\alpha/\alpha$ (%)
Cosmological Redshift z